

Lab Handout: Setting Up a Virtual Machine, Installing Ubuntu and Using Windows Subsystem for Linux

Introduction

Virtualisation allows a *guest* operating system (OS) to run inside a *host* OS by emulating hardware devices. A virtual machine (VM) has its own memory, CPU and storage that are isolated from the host, enabling you to experiment safely. The Windows Subsystem for Linux (WSL) is different: rather than running a full VM, Windows integrates a Linux compatibility layer so that Linux binaries run directly on top of the Windows kernel. WSL therefore starts faster and uses fewer resources, but it does not provide a full hypervisor environment.

This handout guides you through creating a VM, installing Ubuntu, and then installing WSL on Windows. The instructions are sufficiently detailed for a four-hour laboratory session.

Part 1 - Creating a Virtual Machine and Installing Ubuntu

Download Ubuntu ISO

Before creating a VM you need an ISO image of Ubuntu. The GeeksforGeeks installation guide recommends downloading the installer from Ubuntu's official website¹. Use the latest Long-Term Support (LTS) release (Ubuntu 24.04 LTS).

Install VirtualBox/VMware on the Host

VirtualBox is a free hypervisor maintained by Oracle. It runs on Windows, macOS and Linux, supports both AMD and Intel processors, allows multiple

¹<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

VMs and offers seamless window integration². Download and install VirtualBox from <https://www.virtualbox.org/>. Accept the default settings during installation and reboot if prompted.

Create a New VM

1. **Launch VirtualBox** and click **New**³. Give the VM a descriptive name such as “Ubuntu VM”. VirtualBox will auto-select the type when the name contains “Ubuntu”.
2. **Allocate memory**. Set the RAM to at least 2 GB (2048 MB); 4 GB is recommended for smoother performance⁴.
3. **Create a virtual hard disk**. Choose the VDI format and a dynamically allocated disk⁵. Set the disk size to at least 20 GB⁶.

Configure the Boot Medium and Start the VM

1. Select your VM and click **Start**. When prompted, browse to the downloaded ISO and attach it as the virtual optical disk⁷.
2. The VM will boot from the ISO. Select **Try or Install Ubuntu**⁸.

Install Ubuntu in the VM

Follow the installer prompts:

- **Keyboard layout**: choose your preferred layout or accept the default⁹.
- **Installation type**: select *Normal Installation* for a full desktop environment¹⁰.
- **Disk partitioning**: allow the installer to use the entire virtual disk; this will not affect your host system¹¹.
- **Time zone**: select your location to set the correct time¹².

²<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

³<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁴<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁵<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁶<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁷<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁸<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

⁹<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁰<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹¹<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹²<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

- **User account:** specify your name, computer name and a secure password¹³.

The installer copies files to the virtual disk and may take several minutes¹⁴. When prompted, reboot the VM¹⁵, and detach the ISO if necessary.

(Optional) Enable Full-Screen Mode

Ubuntu initially may not fill the VM window. VirtualBox's *Guest Additions* provide better video drivers. In the VM menu choose **Devices** → **Insert Guest Additions CD**. Open the mounted disk inside Ubuntu and run 'autorun.sh' from a terminal¹⁷. Enter your password when asked. After installation, reboot the VM¹⁸; you should now be able to use full-screen mode.

Recommended VM Settings

Setting	Recommendation	Notes
RAM	≥ 2 GB	More memory improves performance
Virtual CPU	2 cores	Assign more cores if the host has them
Storage controller	VDI dynamic	Dynamically allocated disks save space
Disk size	≥ 20 GB	Desktop installation may require more
Network mode	NAT	Provides Internet access through the host

Virtualisation Concept Illustration

Part 2 - Installing Windows Subsystem for Linux (WSL)

Check System Requirements

WSL is supported on Windows 10 version 2004 (build 19041) or later and on Windows 11¹⁹. Earlier versions require a manual installation and only support WSL 1.

¹³<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁴<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁵<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁶<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁷<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁸<https://www.geeksforgeeks.org/linux-unix/how-to-install-ubuntu-on-virtualbox/>

¹⁹<https://learn.microsoft.com/windows/wsl/install>

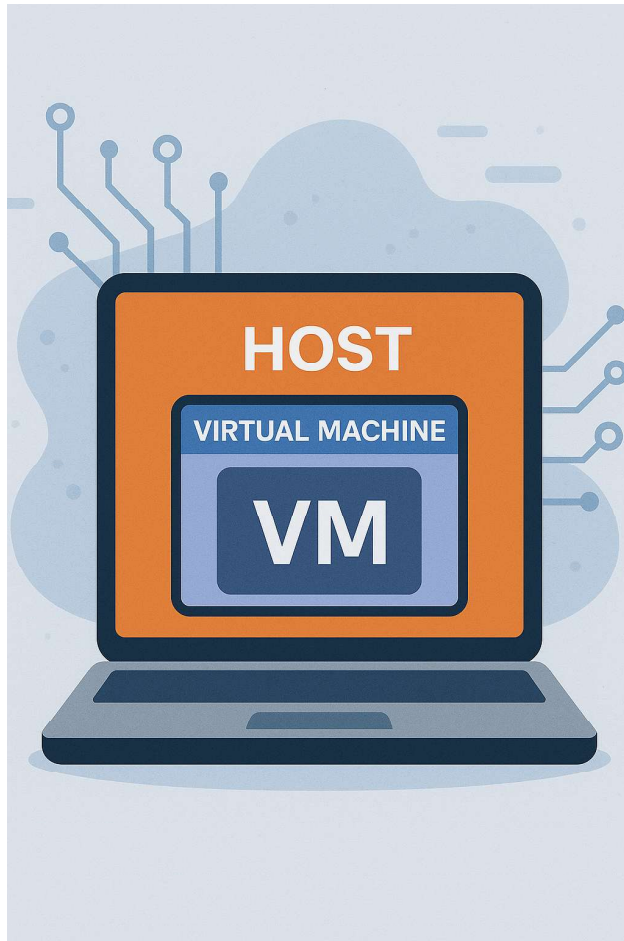


Figure 1: Abstract representation of a host computer running a virtual machine.

Install WSL Using a Single Command

Microsoft has simplified installation with a single command. Open PowerShell or Command Prompt as administrator and run²⁰:

```
wsl --install
```

This command enables the WSL and Virtual Machine Platform features, downloads the Linux kernel, sets WSL 2 as the default version and installs the Ubuntu distribution by default. To install a different distribution, append the '-d' flag (e.g., 'wsl -install -d Debian')²¹. Restart the computer when prompted and create a Linux username and password on the first launch²².

²⁰<https://learn.microsoft.com/windows/wsl/install>

²¹<https://learn.microsoft.com/windows/wsl/install>

²²<https://learn.microsoft.com/windows/wsl/install>

Manual Installation Steps (Older Versions)

If 'wsl -install' is unavailable, follow these steps:

1. **Enable WSL feature:** run the following in an elevated PowerShell²³:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all
```

2. **Enable Virtual Machine Platform:** WSL 2 requires this feature. Execute²⁴:

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

Then reboot.

3. **Install the Linux kernel update:** download the WSL 2 Linux kernel update package from Microsoft and run it²⁵.
4. **Set WSL 2 as default (optional):** run 'wsl -set-default-version 2' to ensure future installations use WSL 2²⁶.
5. **Install a Linux distribution:** use Microsoft Store or run 'wsl -install -d <DistributionName>'²⁷.

Install and Use Linux Packages in WSL

Once WSL is installed, you can update packages and install software similarly to a native Linux system. For example, to install the htop system monitor²⁸:

```
sudo apt update
sudo apt install htop
```

WSL Concept Illustration

Part 3 - Comparing Virtual Machines and WSL

Feature	Virtual Machine (VirtualBox/VMware)	Windows Subsystem for Linux (WSL)
Hardware isolation	Full isolation with dedicated RAM/CPU	Shares Windows kernel; lightweight
Resource usage	Uses more memory and CPU	Minimal overhead
Start-up time	Slower; full OS boot	Instant start (seconds)
Graphics/UI support	Full desktop environment	CLI by default; GUI apps require WSLg
Use case	Testing multiple OSs, GUI apps, kernel development	Running Linux command-line tools on Windows

²³<https://learn.microsoft.com/windows/wsl/install-manual>

²⁴<https://learn.microsoft.com/windows/wsl/install-manual>

²⁵<https://learn.microsoft.com/windows/wsl/install-manual>

²⁶<https://learn.microsoft.com/windows/wsl/install>

²⁷<https://learn.microsoft.com/windows/wsl/install>

²⁸<https://www.geeksforgeeks.org/installation-guide/how-to-install-wsl2-windows-subsystem-for-linux-2->



Figure 2: Abstract illustration of Windows and Linux working together inside WSL.

Conclusion

You have learned how to create a virtual machine and install Ubuntu, and how to enable Windows Subsystem for Linux to run Linux workloads on Windows. VMs provide full isolation and are ideal for experimenting with complete operating systems. WSL is lightweight and tightly integrated with Windows, making it perfect for development workflows requiring Linux tools without leaving Windows. Depending on your needs, you may choose one or use both together.